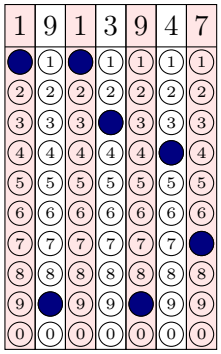
TEST VARIANTI RAQAMI: **1913947**

MATEMATIKA

Ushbu test varianti 30 ta test topshirig‘idan iborat. Uni bajarishga 1 soat vaqt belgilash tavsiya etiladi. Natijani bilish uchun javoblaringizni javoblar varaqasiga belgilab, rasmini **@dtmxizmatbot** orqali jo‘nating. Sizga omad tilaymiz!

TEST VARIANTI RAQAMINI JAVOBLAR VARAQASIGA TO‘G‘RI KO‘CHIRING!



1. To‘g‘ri silindrning balandligi 5 cm va asosining radiusi 4 cm. Uning yon sirtidagi A va B nuqtalar asos tekisligidan mos ravishda 2 va 3 cm balandlikda joylashgan. Agar AB kesmaning uzunligi 5 cm bo‘lsa, silindr o‘qidan AB kesmagacha bo‘lgan eng yaqin masofani (cm) toping.

A) $\sqrt{10}$ B) $\sqrt{11}$ C) $\frac{\sqrt{39}}{2}$ D) $\frac{\sqrt{41}}{2}$

2. Karim ota 71 yoshda. Uning nabiralarining o‘rtacha yoshi 21 da. Nabiralari bilan Karim otaning yoshlari o‘rta arifmetigi 26 ga teng. Karim otaning nechta nabirasi bor?

A) *8* B) *10* C) *12* D) *9*

3. Tenglamani yeching:
$\frac{x+3}{4^2-1}+\frac{x+3}{6^2-1}+\frac{x+3}{8^2-1}+...+\frac{x+3}{100^2-1}=\frac{49}{101}$

A) 2 B) 0 C) -3 D) $-\frac{1}{3}$

4. $\frac{x-25}{5+\sqrt{x}}+\frac{x^2-6}{\sqrt{6}-x}$ ifodaning $x=6$ dagi qiymatini toping.

A) -11 B) $-2\sqrt{6}$ C) 1 D) -1

5. Arifmetik progressiyaning dastlabki sakkizta hadi yig‘indisi 248 ga, dastlabki o‘n to‘rtta hadi yig‘indisi esa 686 ga teng. Arifmetik progressiyaning beshinchi hadini toping.

A) *40* B) *34* C) *36* D) *42*

6. Agar $\lg(x+3)-\lg\frac{1}{x}=1$ bo‘lsa, x ni toping.

A) *5* B) *2* C) -5 D) -2

7. Oy o‘qiga perpendikulyar va $y=(x-4)^2\cdot e^x+2$ funksiya grafigiga urinma bo‘lgan to‘g‘ri chiziq tenglamalarini aniqlang.

A) $y=2\cdot e^2+4;\ y=2$
B) $y=4\cdot e^2+2;\ y=4$
C) $y=2\cdot e^2+4;\ y=4$
D) $y=4\cdot e^2+2;\ y=2$

8. 0, 1, 2, 3, 4, 5 raqamlardan jami nechta 3 xonali sonlar tuzish mumkin?

A) *210* B) *180* C) *216* D) *125*

9. $\vec{a}(1;2)$; $\vec{b}(2;1)$ va $\vec{c}(x;2)$ vektorlar uchun $\vec{a}\cdot\vec{c}+\vec{b}\cdot\vec{c}+\vec{a}\cdot\vec{b}=13$ bo‘lsa, x ni toping.

A) *2* B) -1 C) -2 D) *1*

10. $\frac{\sqrt{2}-15}{\sqrt{\sqrt{2}+1}-4}-\sqrt{\sqrt{2}+1}$ ni hisoblang.

A) *4* B) -2 C) *2* D) *0*

11. Agar $f\left(\frac{2x-1}{3}\right)=\frac{6x+5}{3}$ bo‘lsa, $f(1)-f(-2)$ ni hisoblang.

A) *9* B) *3* C) -12 D) -3

12. A murakkab sonlar to‘plami va B juft sonlar to‘plami bo‘lsa, $A\cap B$ to‘plamni aniqlang.

A) $\varnothing$ B) $\{x|x=2n,\ n\in N\}$
C) $\{x|x=2n+2,\ n\in N\}$
D) $\{x|x=2n-2,\ n\in N\}$

13. $7^3+11\cdot 7$ ni 6 ga bo‘lgandagi qoldiqni toping.

A) *2* B) *1* C) *3* D) *0*

14. Hisoblang: $\frac{16-0,36^2}{1,4\cdot 4,12-1,52\cdot 1,4}$.

A) *3,64* B) *2,36* C) *2,6* D) *4,36*

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15. ABC uchburchakning burchaklari 2:3:1 nisbatda. Agar eng kichik tomoni 5 cm bo‘lsa, eng katta tomoni uzunligini (cm) toping.
A) $5\sqrt{3}$ B) 10 C) $5\sqrt{2}$ D) 8

16. $f(x)=\dfrac{17}{3x^2}+\dfrac{75x^2}{17}$ funksiyaning eng kichik qiymatini toping.
A) *8* B) *10* C) *6* D) *12*

17. $(5a-3b)-(-4b+7a)$ ifodani soddalashtiring.
A) $2a+b$ B) $-2a+b$ C) $2a-b$
D) $-2a-b$

18. Agar $a=\dfrac{2}{3}$ bo‘lsa,
$\dfrac{a+4}{a^2-5a}\cdot\sqrt{\dfrac{a^2-10a+25}{a^2+8a+16}}-2\dfrac{1}{2}$ ifodaning qiymatini toping.
A) $-1\dfrac{1}{2}$ B) -1 C) $-2\dfrac{1}{2}$ D) -4

19. $y=\sqrt{2^{2x}-3\cdot 2^{x+1}-16}$ funksiyaning aniqlanish sohasini toping.
A) $x\geq 2$ B) $x\leq 1, x\geq 4$ C) $x\geq 3$
D) $x\leq 2, x\geq 3$

20. $ABCD$ parallelogramda C o‘tkir burchak. E nuqta AB tomonda yotadi. Agar AE:EB nisbat 2:3 kabi bo‘lsa, $AECD$ to‘rtburchak yuzining BCE uchburchak yuziga nisbatini toping.
A) $\dfrac{7}{3}$ B) $\dfrac{5}{3}$ C) $\dfrac{3}{2}$ D) $\dfrac{4}{3}$

21. Koordinata o‘qlarining (3; 0) va (0; 4) nuqtalaridan o‘tadigan chiziqli fuksiyani toping.
A) $y=-\dfrac{4}{3}x+4$ B) $y=-\dfrac{4}{3}x-4$
C) $y=\dfrac{4}{3}x+4$ D) $y=\dfrac{4}{3}x-4$

22. Tenglamani yeching: $tg(4x+\pi)\cdot tg(3x)=1$
A) $x=\dfrac{\pi}{14}+\dfrac{\pi k}{7},\ k\in Z$
B) $x=\dfrac{\pi}{4}+\dfrac{\pi k}{7},\ k\in Z$
C) $x=\dfrac{\pi}{3}+\dfrac{\pi k}{7},\ k\in Z$
D) $x=\dfrac{\pi}{7}+\dfrac{\pi k}{7},\ k\in Z$

23. Detal uzunligi 1:8 masshtabli xaritada 4,5 cm bo‘lsa, 1:5 masshtabli xaritada necha santimetr bo‘ladi?
A) *6,8* B) *7,8* C) *7,2* D) *7*

24. $\begin{cases} 6x>x^2 \\ 4x^2\leq 25 \end{cases}$ tengsizliklar sistemasining butun yechimlari yig‘indisini toping.
A) *4* B) *3* C) *7* D) *12*

25. $2x^2+\left(6-\sqrt{3}\right)x=3\sqrt{3}$ tenglamaning eng katta ildizini toping.
A) $0,5\sqrt{3}$ B) $2\sqrt{3}$ C) $\sqrt{3}$ D) 3

26. Hisoblang: $\dfrac{111}{333}+\dfrac{222}{666}+\dfrac{333}{999}$
A) *2* B) *1* C) *1,6* D) *1,5*

27. $2\cos^2 x-\cos x-1=0$ tenglamaning eng kichik musbat yechimini toping.
A) $\dfrac{3\pi}{4}$ B) $\dfrac{\pi}{4}$ C) $\dfrac{\pi}{6}$ D) $\dfrac{2\pi}{3}$

28. α tekislik va uni kesib o‘tadigan AB kesma berilgan. Kesmaning uchlaridan α tekislikkacha bo‘lgan masofalar AA_1=12 cm, BB_1=13 cm bo‘lsa, AB kesmani A uchidan boshlab hisoblaganda 3:2 nisbatda bo‘luvchi C nuqtadan α tekislikkacha bo‘lgan masofani (cm) toping.
A) *4* B) *3,6* C) *3* D) *3,2*

29. $\int(2x+1)\cos(x^2+x)dx$ integralni hisoblang.
A) $\cos(x^2+x)+C$ B) $\sin(x^2+x)+C$
C) $-\sin(x^2+x)+C$ D) $-\cos(x^2+x)+C$

30. Uchburchakning asosiga tushirilgan balandligi 20 cm bo‘lib, asosidan 2 cm ga katta. Agar shu balandlik 10% ga kamaytirilib, asosi esa 4 cm ga uzaytirilsa, uchburchakning yuzi qanday o‘zgaradi?
A) *12 cm^2ga ortadi* B) *18 cm^2ga kamayadi*
C) *18 cm^2ga ortadi* D) *o‘zgarmaydi*

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Test varianti "Test topshiriqlari to'plami 2019" asosida shakllantirilgan:

1 – 113-bet 54	7 – 83-bet 52	13 – 8-bet 20	19 – 52-bet 1	25 – 62-bet 29
2 – 21-bet 43	8 – 122-bet 2	14 – 13-bet 39	20 – 102-bet 89	26 – 11-bet 14
3 – 63-bet 44	9 – 117-bet 24	15 – 90-bet 6	21 – 71-bet 16	27 – 43-bet 80
4 – 25-bet 31	10 – 25-bet 34	16 – 78-bet 13	22 – 37-bet 28	28 – 108-bet 16
5 – 33-bet 32	11 – 76-bet 67	17 – 45-bet 6	23 – 19-bet 30	29 – 85-bet 8
6 – 58-bet 29	12 – 121-bet 17	18 – 50-bet 48	24 – 68-bet 19	30 – 98-bet 63

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